

Program : Diploma in Fire Technology and Safety	
Course Code: 4468	Course Title: Fluid Mechanics Lab
Semester: 4	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To apply fluid mechanics principles to fire safety applications.
- To familiarize students with fire water supply systems and pressure measurement techniques.
- To analyze fluid flow in pipes and firefighting equipment.

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Measure pressure and determine metacentric height of floating bodies.	9	Applying
CO2	Analyze fluid flow properties, verify Bernoulli's theorem, and assess firefighting water supply systems.	10	Applying
CO3	Evaluate pipe fittings, friction losses, and minor losses in fire hose systems.	10	Applying
CO4	Determine discharge coefficients of fire hydrants and sprinklers.	12	Applying
	Lab Exam	4	

CO–PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						1
CO2				2			1
CO3			2				1
CO4							3

3-Stronglymapped,2-Moderatelymapped,1-Weaklymapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	CognitiveLevel
CO1	Measure pressure and determine metacentric height of floating bodies.		
M1.01	Study of pressure measuring devices (manometer, Bourdon gauge)	6	Understanding
M1.02	Determination of metacentric height of floating body	3	Applying
CO2	Analyze fluid flow properties, verify Bernoulli's theorem, and assess firefighting water supply systems.		
M2.01	Verification of Bernoulli's theorem	4	Applying
M2.02	Measurement of velocity and discharge using venturi meter	6	Applying
	Lab Exam I	2	
CO3	Evaluate pipe fittings, friction losses, and minor losses in fire hose systems.		
M3.01	Study of firefighting pipe fittings, joints, and valves	4	Understanding
M3.02	Determination of friction losses in fire hoses	3	Applying
M3.03	Measurement of minor losses in pipe flow	3	Applying
CO4	Determine discharge coefficients of fire hydrants and sprinklers.		
M4.01	Determination of discharge coefficient of hydrants	6	Applying

M4.02	Study of sprinkler discharge patterns and performance	6	Applying
	LabExam–II	2	